

AI Interview Studio

Technical Documentation

1. Technology Stack

The AI Interview Studio is built upon a modern, full-stack architecture prioritizing low latency, sophisticated UI, and robust AI integration.

Frontend Layer

- **React.js (v19)**: Core UI engine for component-based architecture and state management.
- **Vite (v8)**: Modern, optimized build tool with Fast Refresh (HMR).
- **Framer Motion**: Animation library for fluid dashboard transitions and micro-interactions.
- **CodeMirror**: Advanced browser-based code editor for programming assessments.
- **Vanilla CSS (Glassmorphism)**: Custom styles featuring transparent background aesthetics and modern neon accents.

Intelligence & Vision Layer

- **Face-API.js**: Browser-based real-time face detection, tracking, and engagement analysis.
- **Voice Activity Detection (VAD)**: Real-time audio buffer analysis to detect whispering or external assistance.
- **OpenRouter (GPT-4o-mini)**: The AI core generating adaptive technical questions and performing semantic grading of responses.

Backend & Data Layer

- **Node.js (LTS) & Express.js (v5)**: Server environment for RESTful APIs, authentication, and logic handling.

- **SQLite3:** Lightweight, local relational database for storing user profiles and assessment histories.
- **Nodemailer:** Automated email dispatch for registrations and performance reports.

2. System Requirements

Hardware Requirements

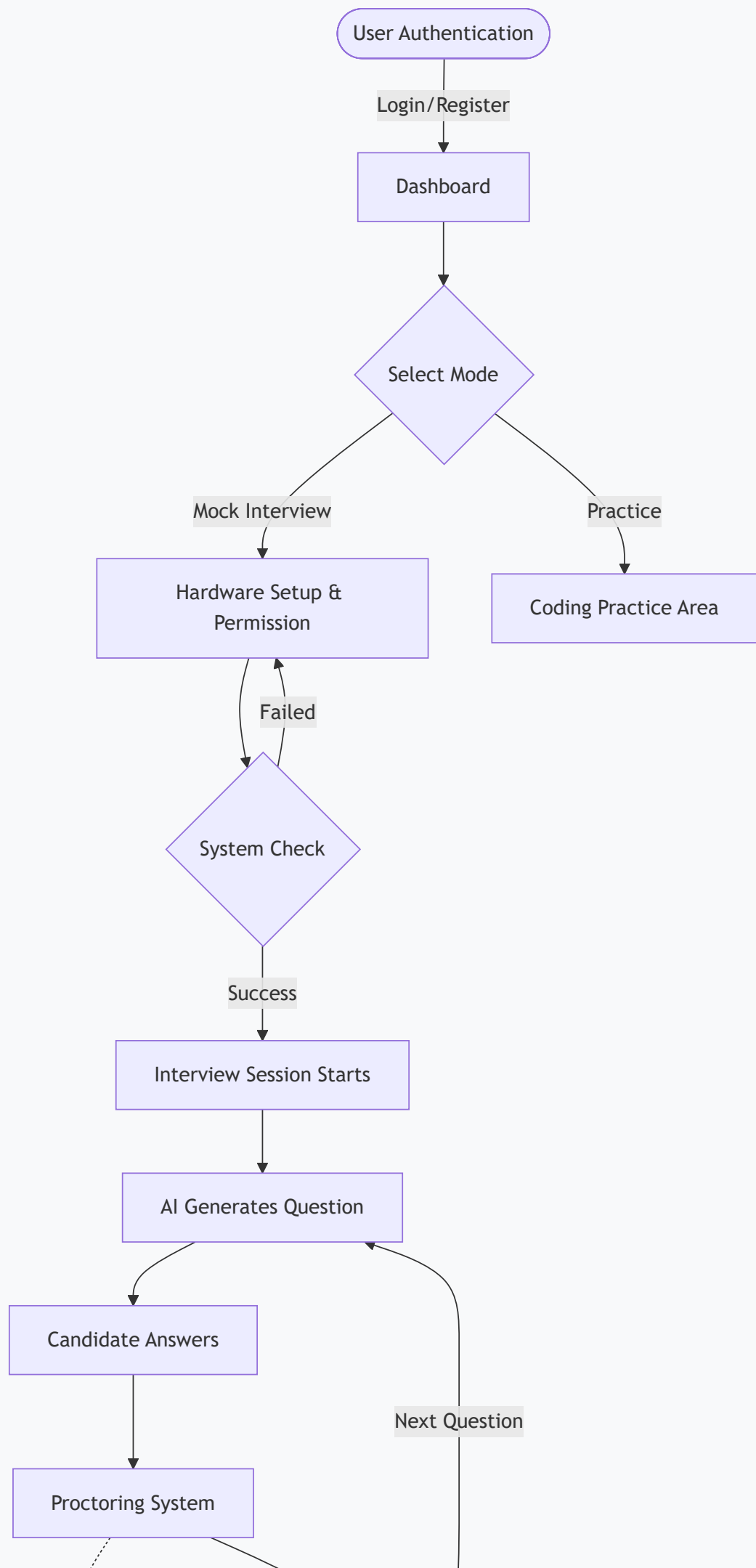
- **CPU:** Dual-core processor (Intel Core i3 / AMD Ryzen 3 or higher).
- **RAM:** 8 GB recommended (Minimum 4 GB) for seamless AI model execution and browser performance.
- **Storage:** 500 MB of available disk space.
- **Peripherals:** Functioning Webcam (720p minimum) and Microphone for live proctoring and voice interaction.

Software Requirements

- **Operating System:** Windows 10/11, macOS, or modern Linux distributions.
- **Web Browser:** Latest versions of Google Chrome, Microsoft Edge, or Mozilla Firefox (Requires support for WebRTC, Canvas API, and WebGL).
- **Environment (Local Setup):** Node.js (v18+) and npm/yarn.

3. Present Flow Diagram

The following diagram illustrates the user journey from registration through the interview process to final assessment.



Anomaly Detected

Proctor Warning /
Terminate

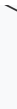
AI Evaluates Response

Session Complete

Generate Score & Feedback

Performance Dashboard

Download Certificate /
Report



4. Methodology Diagram

This block diagram details the interaction between the system's modular layers during an active assessment.

